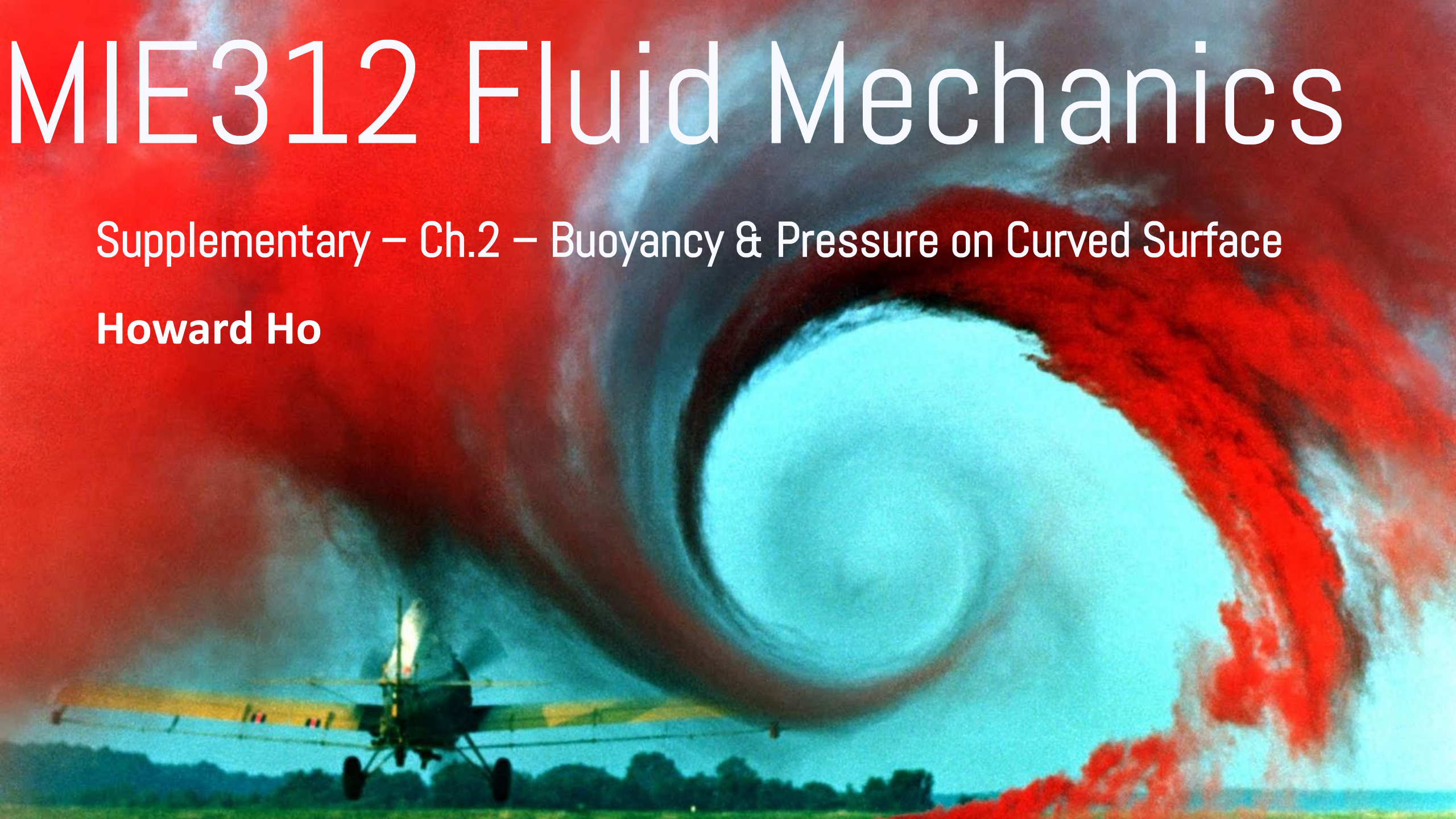
MIE312 Fluid Mechanics

Supplementary – Ch.2 – Buoyancy & Pressure on Curved Surface

Howard Ho



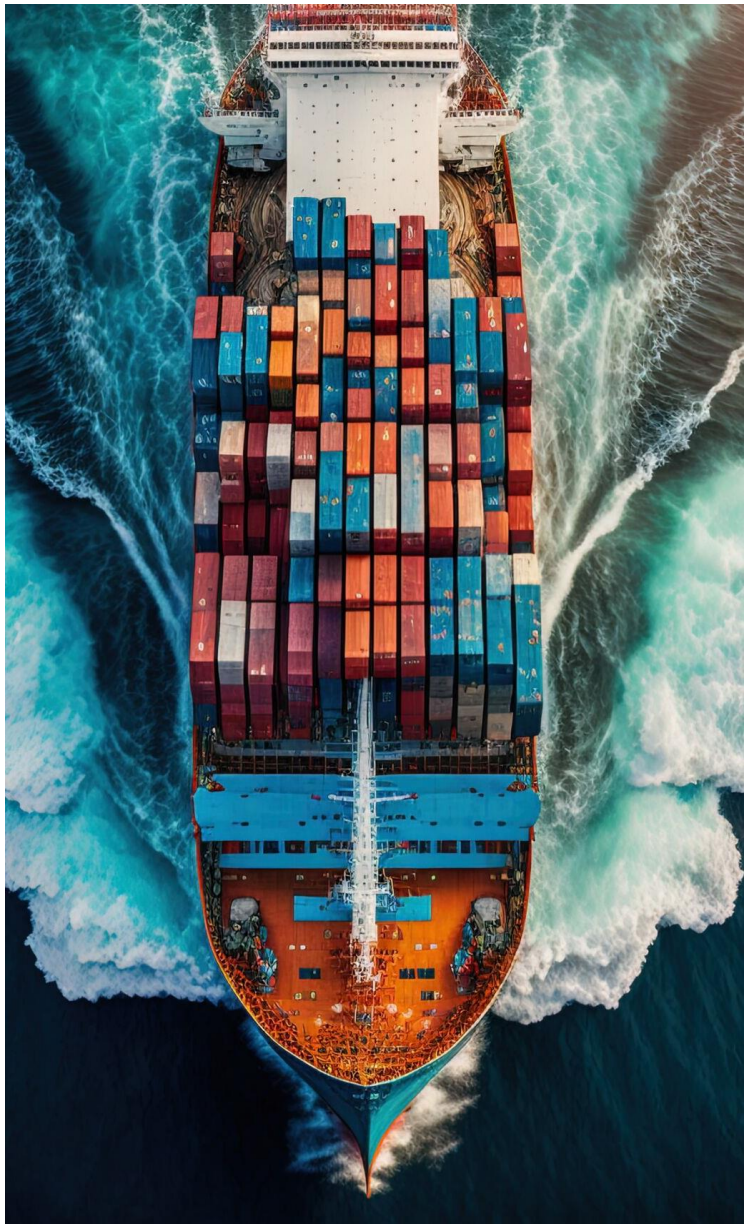
Learning Objective

- Concept of buoyancy
- Calculate buoyancy forces on partially submerged objects
- Calculate hydrostatic pressure on curved surface

Link to recording:
https://youtu.be/MTn-IvcDZ14

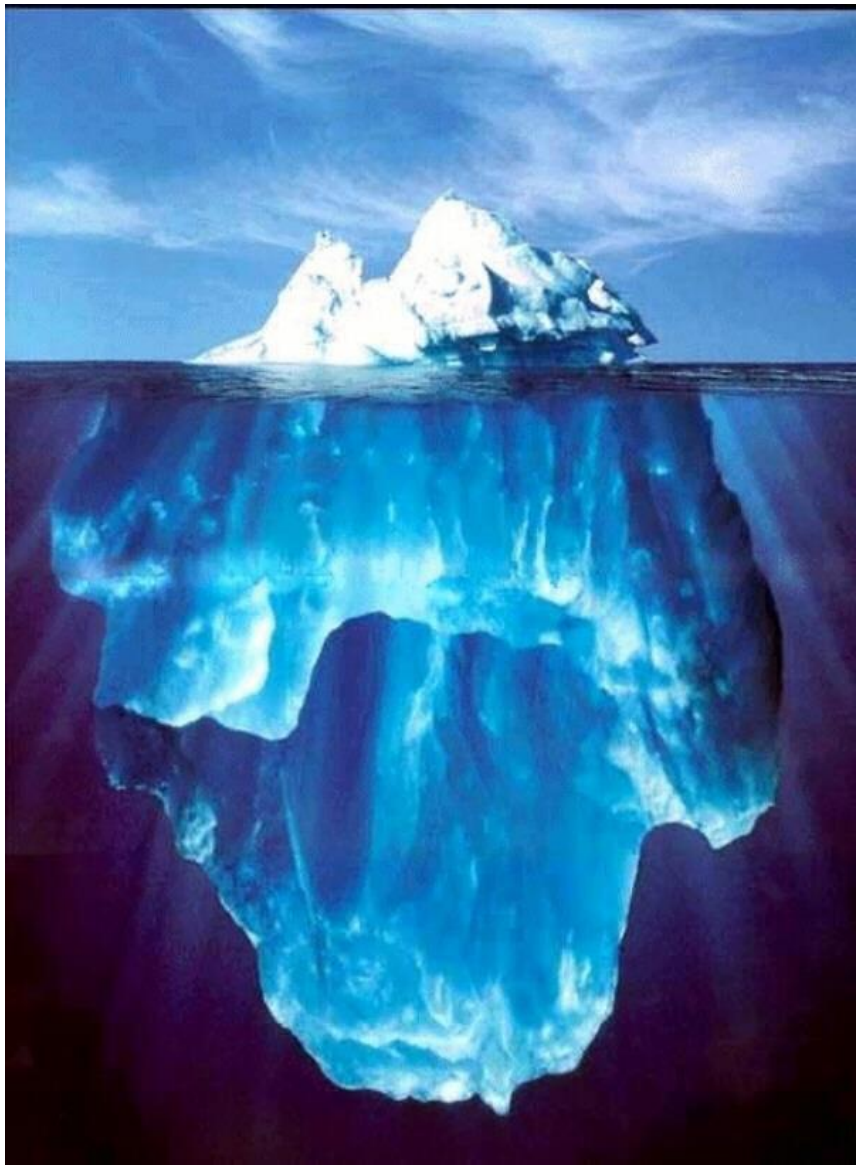


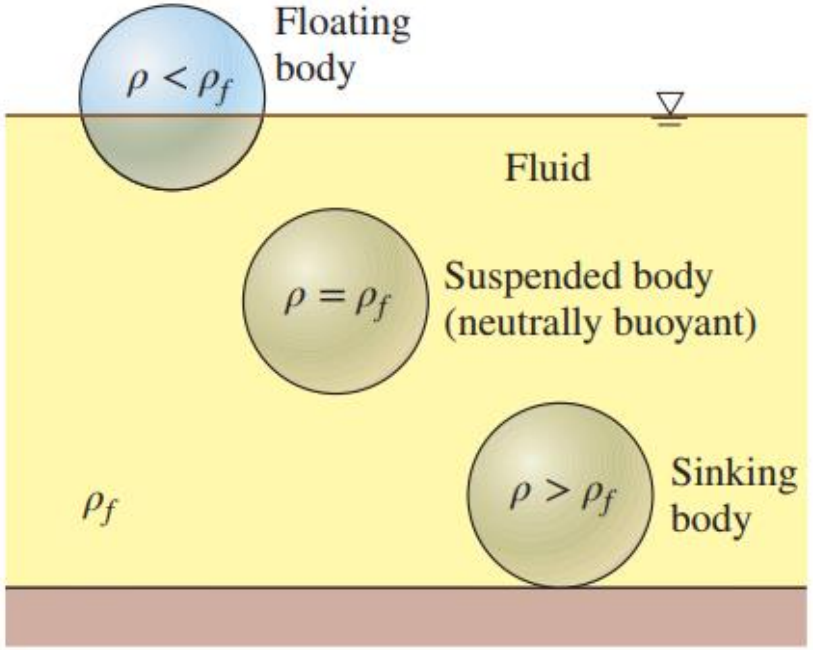


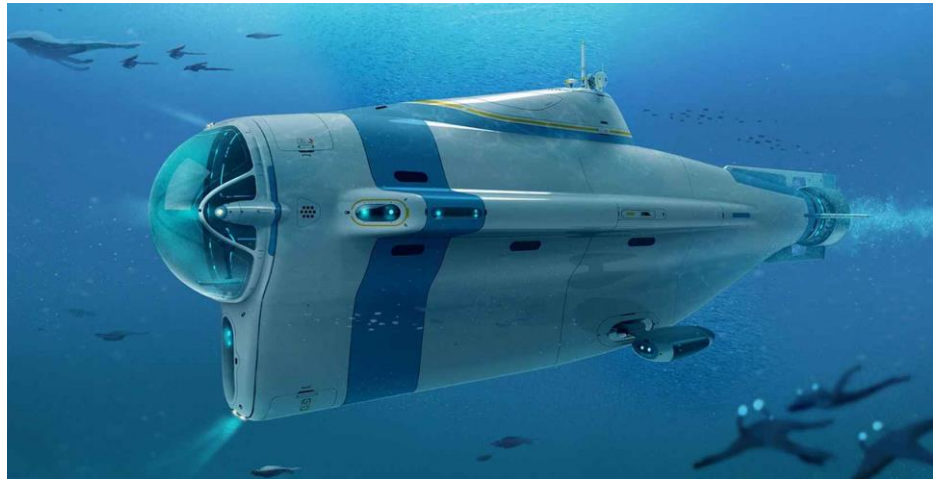


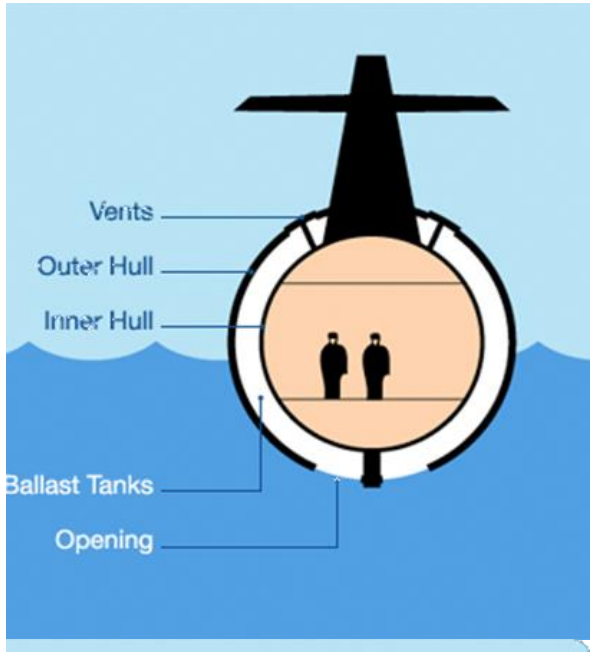
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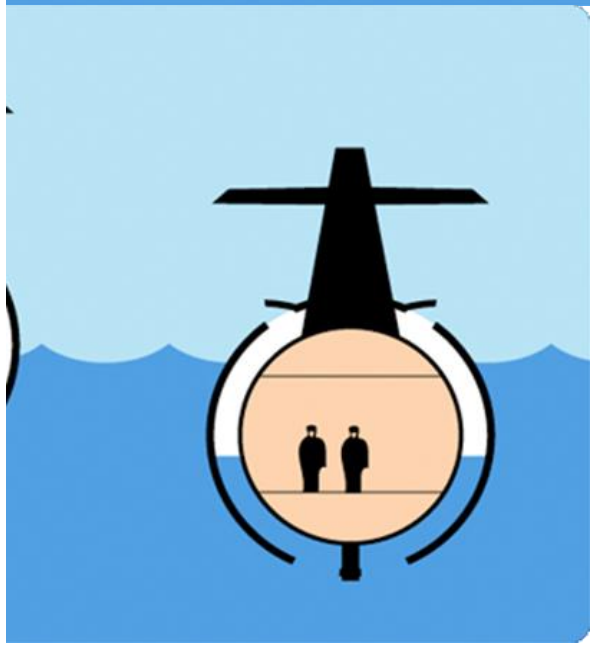
Buoyancy











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Buoyancy

- A fully submerged object experiences pressure on all sides
- Recall pressure force $F_P = \rho g h \times Area$
- If the object has thickness, $F_{P,\,lower} > F_{P,\,upper}$

$$F_{Buoyancy} = F_{P,\,lower} - F_{p,\,upper}$$

$$F_B = \rho g h_2 A - \rho g h_1 A$$

$$F_B = \rho_f g \forall_{sub}$$

- F_B acts on the centroid of the submerged volume $\forall_{sub}$
- If object is suspended (not moving):

$$F_B = W_{object}$$

$$\rho_f g \forall_{sub} = \rho_{object} g \forall_{total}$$

$$\frac{\forall_{sub}}{\forall_{total}} = \frac{\rho_{object}}{\rho_f}$$

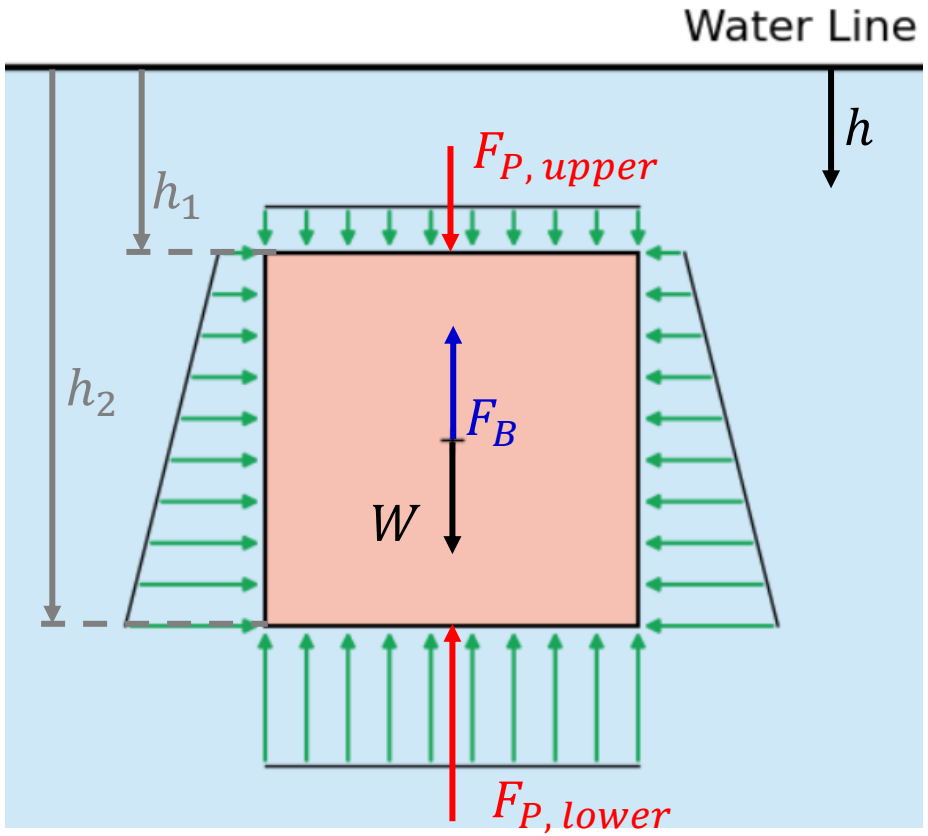
ρ_f = Fluid Density

$\forall_{sub}$= Submerged Volume

W = weight of object

*Avoid using lower case v for volume

We get to ignore the atmospheric pressure's effect because they will get cancelled out



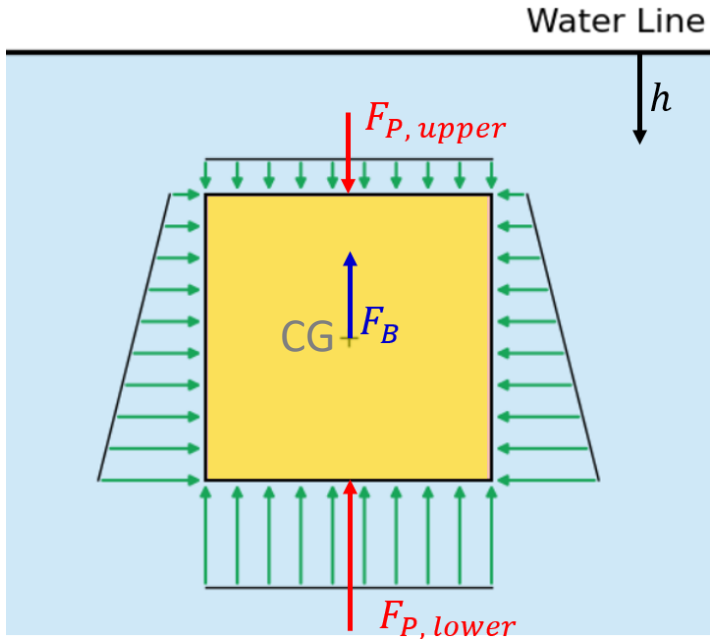
Green arrows = pressure distribution

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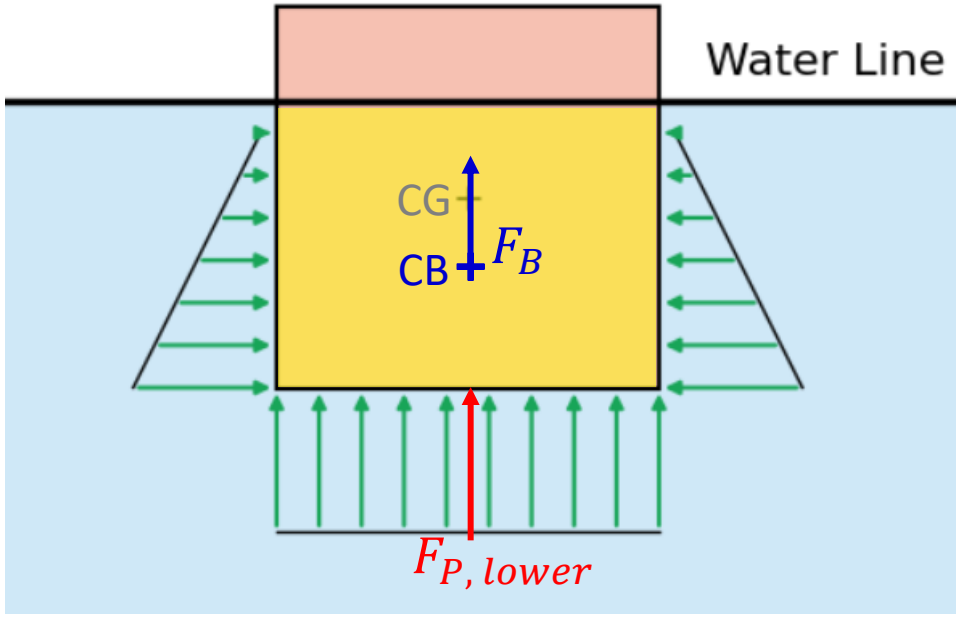
Partially Submerged

- When object is only partially submerged, adjustment is needed
- Upper surface pressure varies
  - If exposed to air: $P_{upper} = P_{atm}$
  - If submerged under a different liquid: $P_{upper} \neq P_{atm}$

$$F_B = \rho_f g \forall_{submerged}$$



- Note the Submerged Volume (yellow portion) has a new centroid (blue cross)
- F_B acts through Centre of Buoyancy (CB)



The weight of the object is not shown here, but it still exists (I omitted it to make the FBD cleaner), it stays constant no matter the submerged portion (unless we are working with a submarine)

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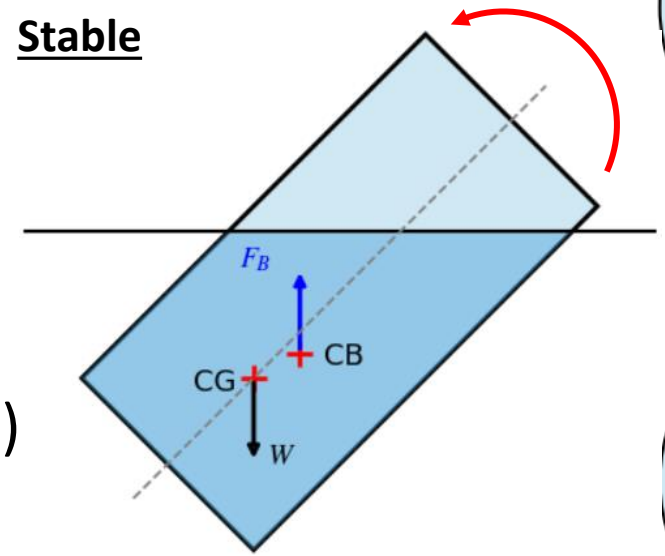
Stability

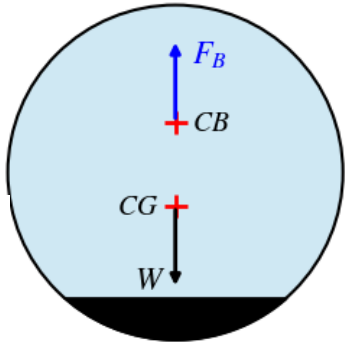
- A moment is present when the object is tilted, this can be:
  - Restoring moment (Stable)
  - Overturning moment (Unstable)
- Depends on the relative locations between CG and CB
- Can be determined through Metacentre (M)



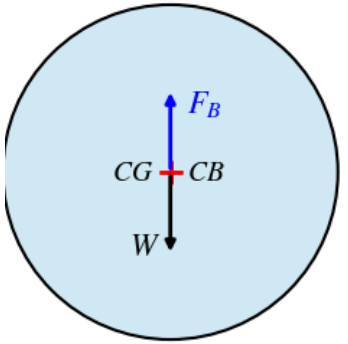
Costa Concordia off the western coast of Italy on January 14, 2012.

Stable



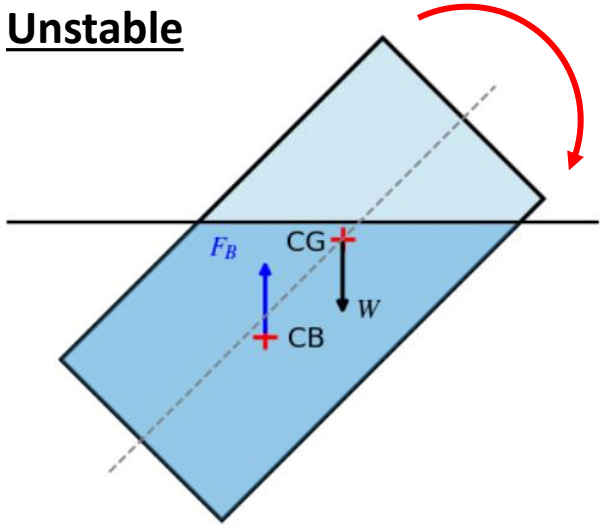


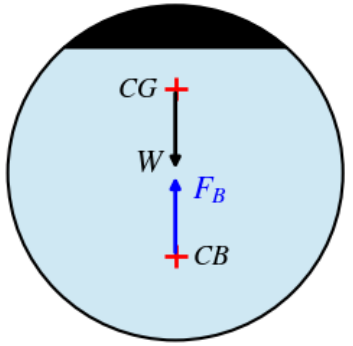
(a) Stable - Bottom heavy



(b) Neutrally Stable

Unstable





(c) Unstable - Top heavy

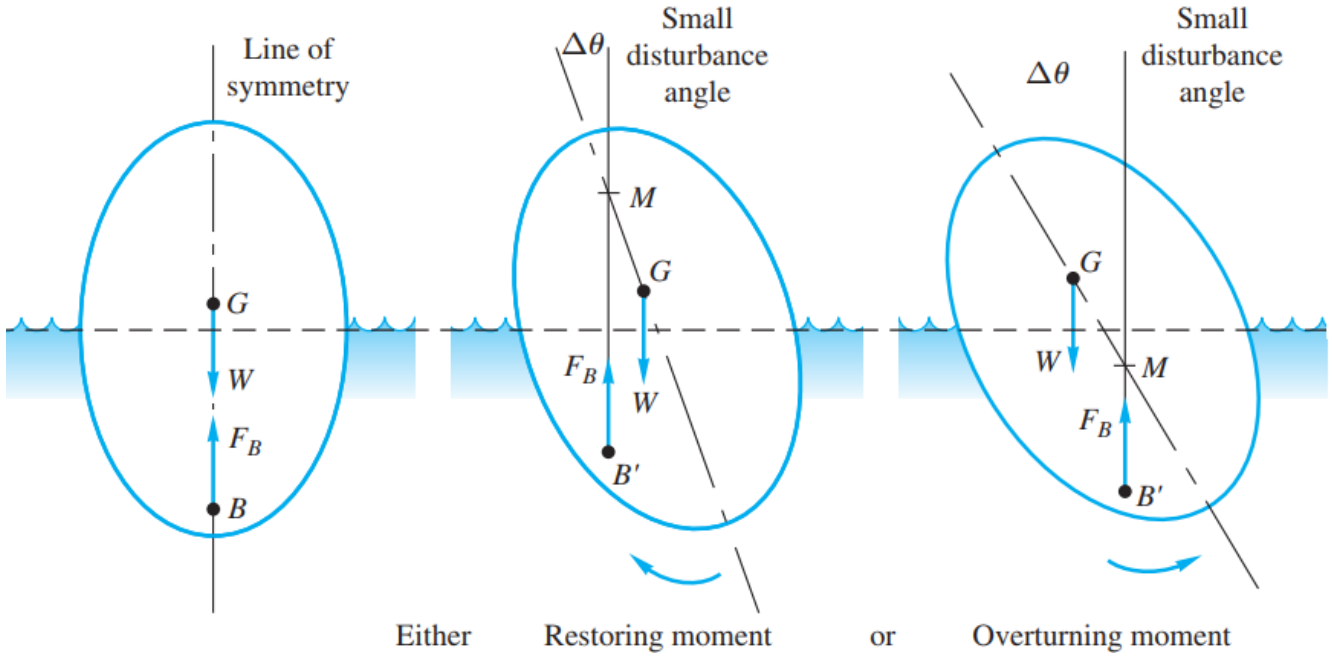
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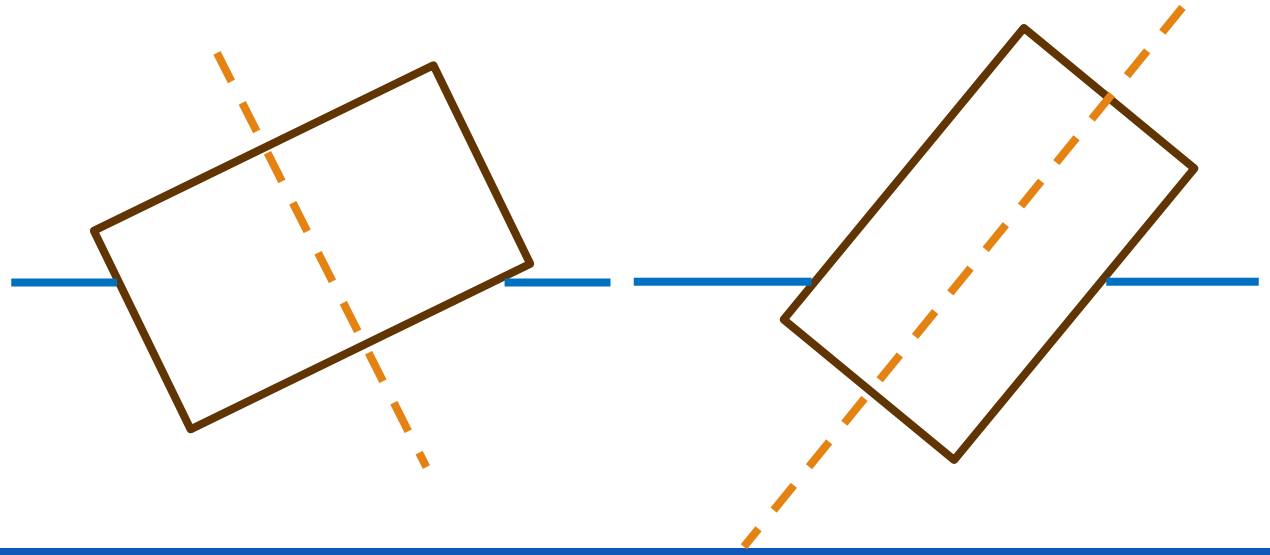
Metacentre

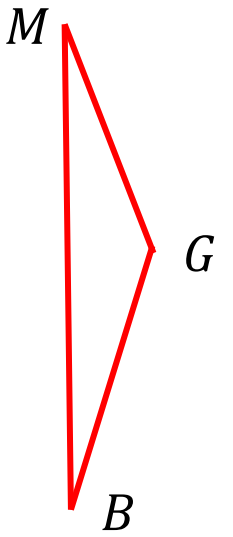
- The Metacentre equation:

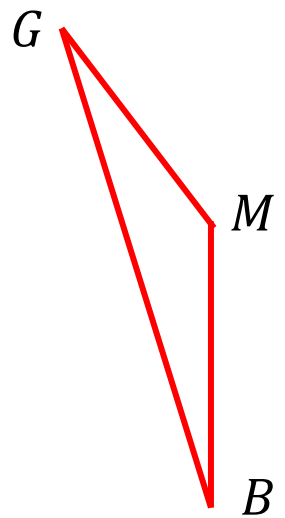
$$\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$$

- $\overline{MG} > 0$ Stable
- $\overline{MG} < 0$ Unstable !







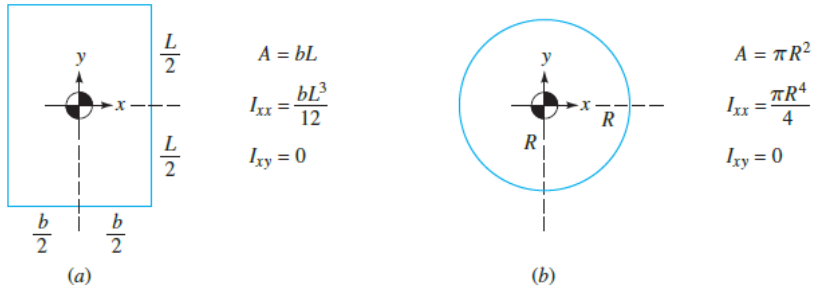


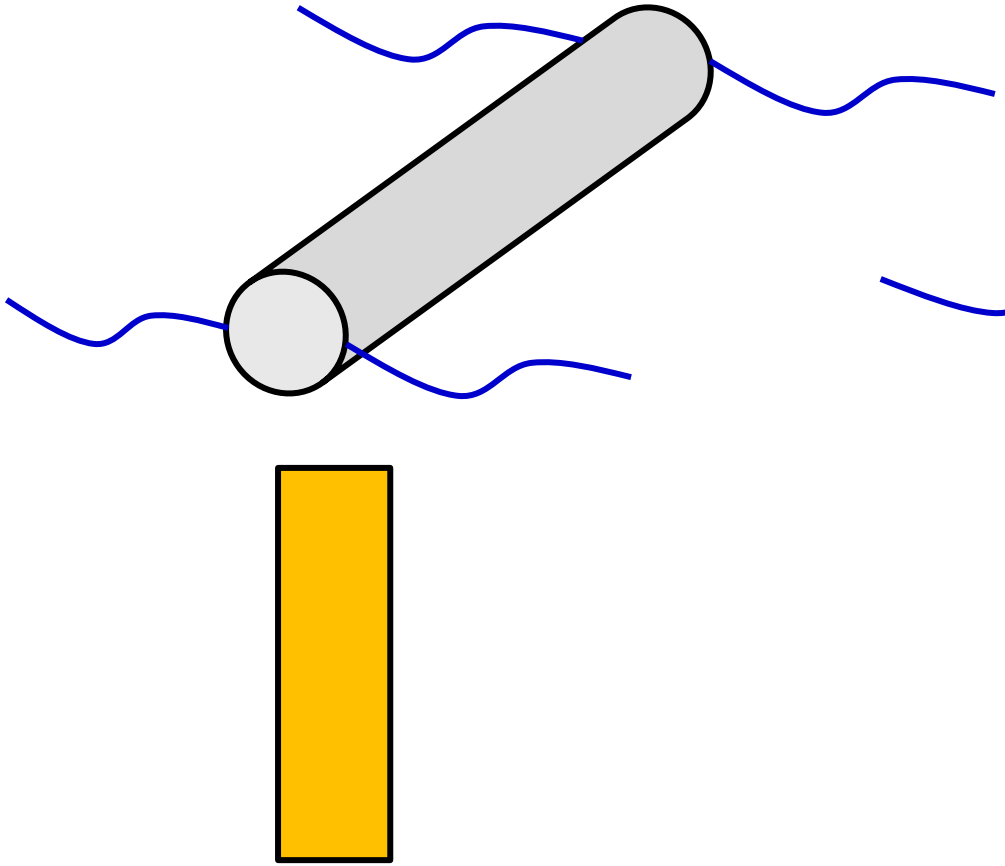
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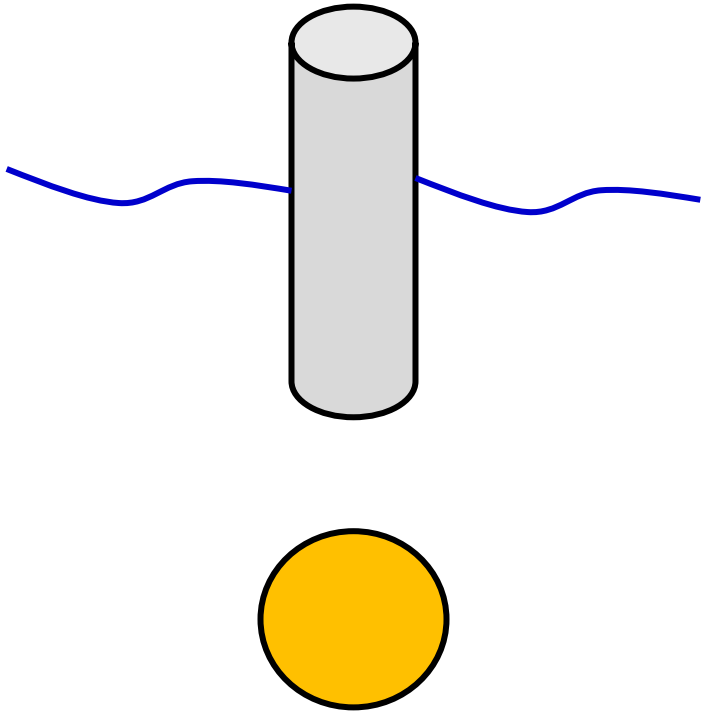
Which Plane for I_o?

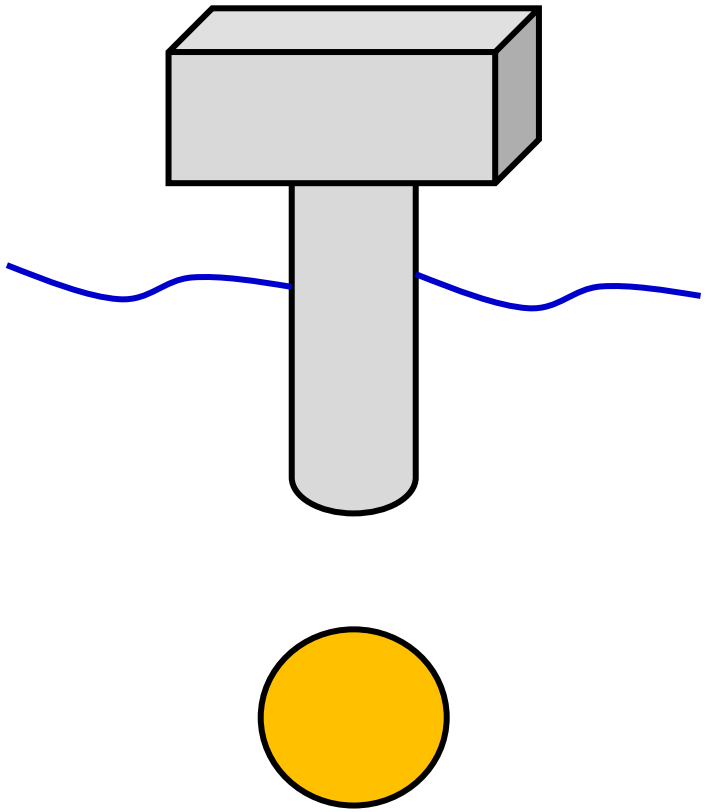
$$\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$$

Note: I_o here is based on the top-down view at water line





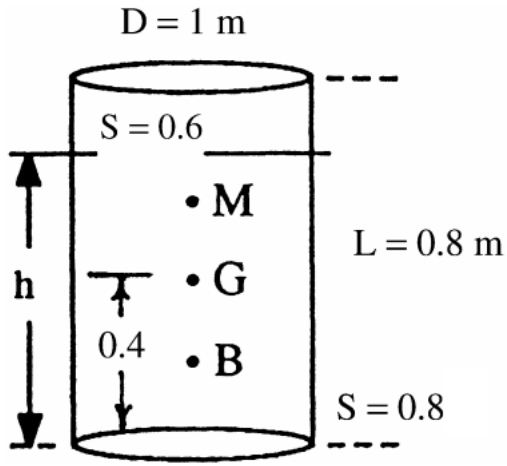




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Problem 1

Consider a wooden cylinder (SG = 0.6) 1 m in diameter and 0.8 m long. Would this cylinder be stable if placed to float with its axis vertical in oil (SG = 0.8)?



2. HYDROSTATICS & ATMOSPHERE

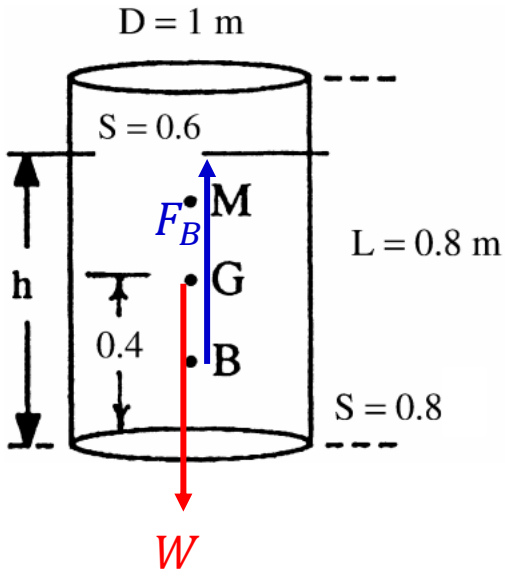
Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$
Isothermal: $P = P_1 \exp[-g\Delta z/RT]$
Troposphere: $P = P_0(1 - \beta z/T_0)^{g/R\beta}$ ($\beta = 0.0065$ K/m)
Hydrostatic: $P = P_0 + \rho gh$
Pascal Law: $F = P \times A$; $P_1 = P_2$
Planes: $F = \gamma h_{CG} A$; $y_{CP} = \frac{I_{xx}\sin\theta}{h_{CG}A}$
Curved: $F_H = \gamma h_{CG} A_{proj}$; F_V = Weight of fluid above
Buoyancy: $F_B = \rho g \forall_{sub}$
Stability: $\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$ (Stable if $\overline{MG} > 0$)
Rigid Body: $\nabla P = \rho(\vec{g} - \vec{a})$; **Rotate:** $z = \frac{\omega^2 r^2}{2g} + C$
Moving manometer: $A_1 a = A_2 b$; inclined-leg vertical change $= b\sin\theta$; write initial/final pressure walks and subtract.

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Problem 1

Consider a wooden cylinder (SG = 0.6) 1 m in diameter and 0.8 m long. Would this cylinder be stable if placed to float with its axis vertical in oil (SG = 0.8)?

$$\rho_{water} = 1025\ kg/m^3$$
$$\rho_{solid} = 0.6\ \rho_{water}$$
$$\rho_{oil} = 0.8\ \rho_{water}$$



$$\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$$

$$I_o = \frac{\pi R^4}{4}, \forall_{sub} = \pi R^2 h$$

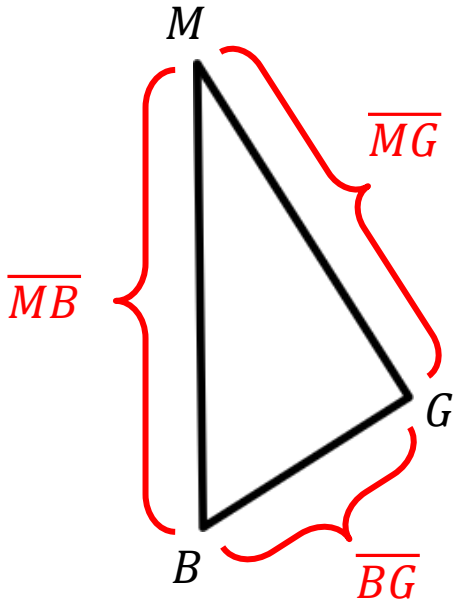
First, find h

$$W = F_B$$

$$\rho_s g \forall_{total} = \rho_{oil} g \forall_{sub}$$

$$0.6(1025)(\pi R^2 L) = 0.8(1025)(\pi R^2 h)$$

$$h = 0.6\ m$$



Point B is @ h/2 = 0.3 m

$$\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$$

$$= \underbrace{\frac{\pi R^4}{4}}_{I_o} \underbrace{\frac{1}{\pi R^2 h}}_{1/\forall_{sub}} - 0.1$$

$$= \frac{R^2}{4h} - 0.1$$

$$= 4.17 \times 10^{-3}\ m$$

$$> 0 \therefore Stable\ ☺$$

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$
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Hydrostatic: $P = P_0 + \rho g h$
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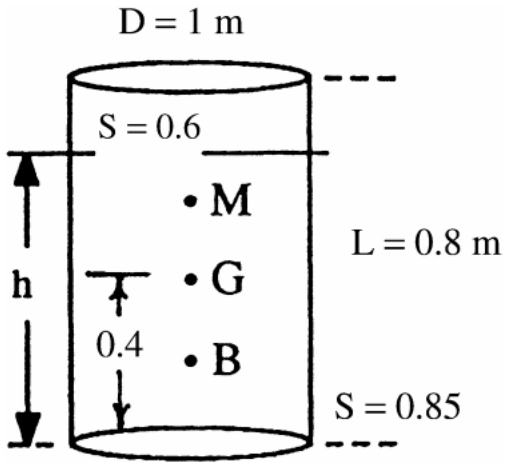
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Problem 1

Consider a wooden cylinder (SG = 0.6) 1 m in diameter and 0.8 m long. Would this cylinder be stable if placed to float with its axis vertical in oil (SG = 0.8)?

SG=0.85

Note: this typed out solution uses SG=0.85 for oil, the method does not deviate from the hand-written version, but the system ends up unstable in the end



Solution: A vertical force balance gives

$$0.85\pi R^2 h = 0.6\pi R^2 (0.8\text{ m}),$$

or: $h = 0.565$ m

The point B is at h/2 = 0.282 m above the bottom. Use Eq. (2.52) to predict the meta-center location:

$$MB = I_o/\upsilon_{sub} = [\pi(0.5)^4/4]/[\pi(0.5)^2(0.565)] = 0.111\text{ m} = MG + GB$$

Now GB = 0.4 m – 0.282 m = 0.118 m, hence **MG** = 0.111 – 0.118 = **–0.007 m**.
This float position is thus **slightly unstable**. The cylinder would turn over. *Ans.*

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Pressure on Curved Surface

- To calculate pressure on curved surface, we can treat it as "steps"*
- Breaking P into horizontal P_x and vertical P_y components
- For F_x, we treat it as a vertical plate:

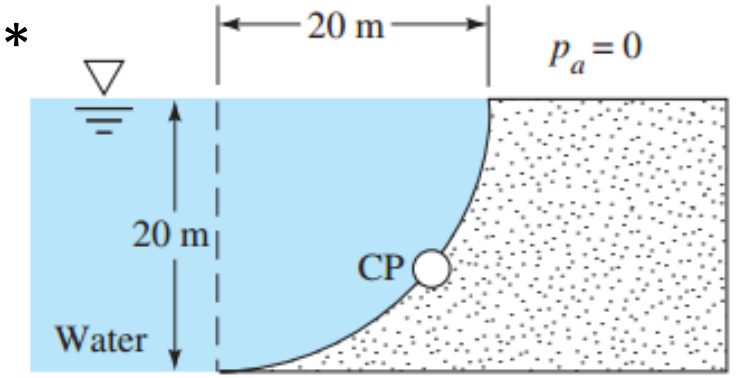
$$F_x = P_x \times A_{vertical} = \rho g h_{CG} A_{vertical}$$

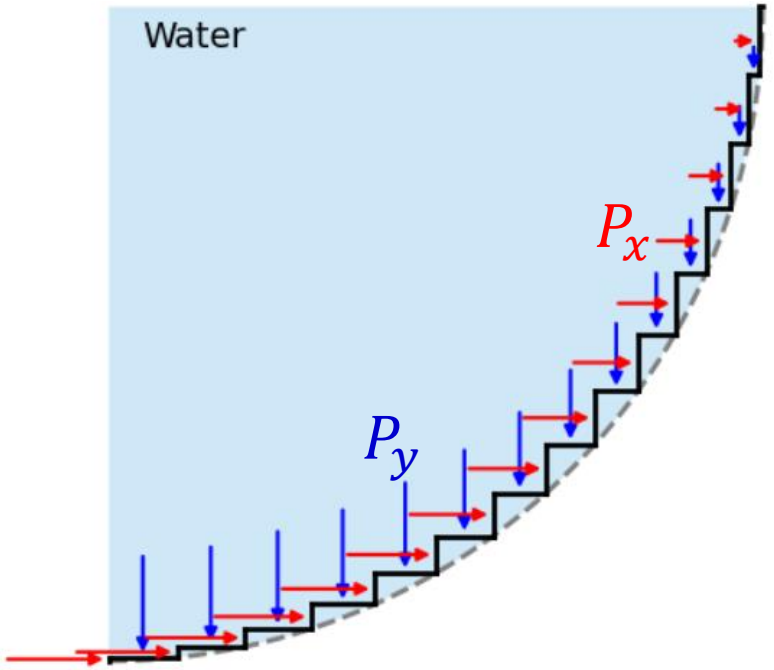
- For F_y, we use the weight of water above:

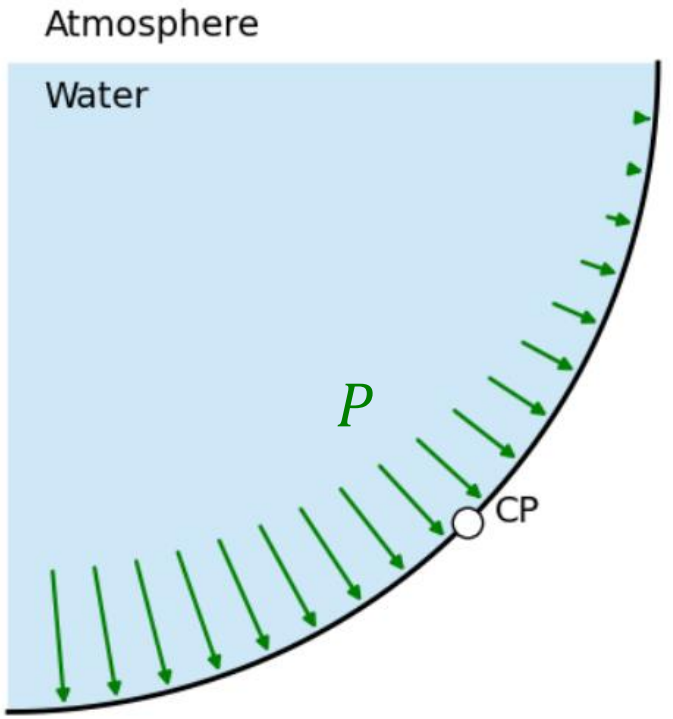
$$F_y = W$$

$$F_y = \rho g \forall_{above}$$

- Use F_x and F_y to find CP



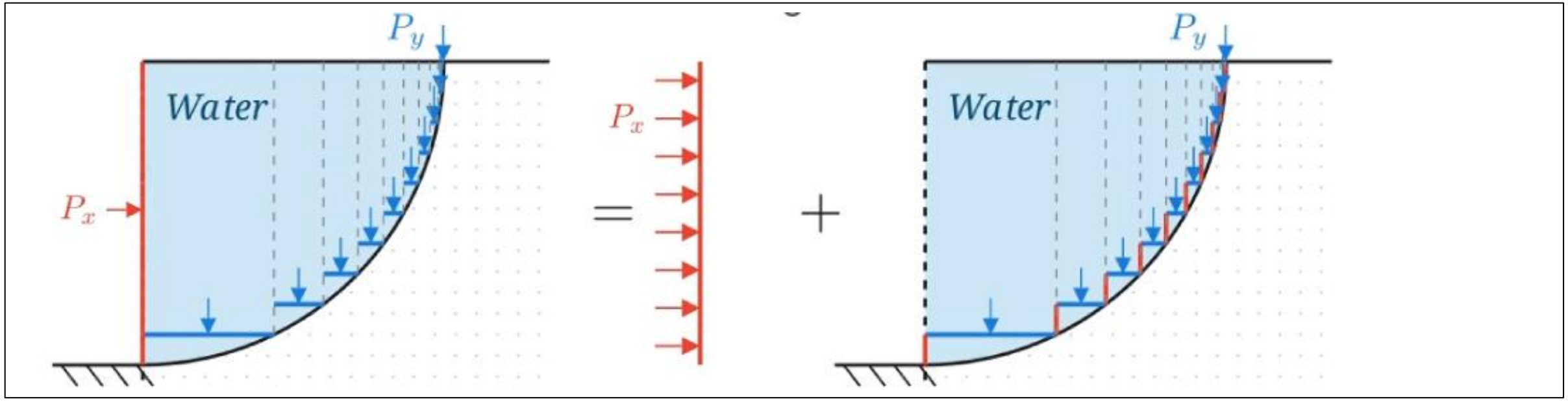




Green arrows: pressure magnitude P
Red arrows: horizontal pressure P_x
Blue arrows: vertical pressure P_y
*It's lowkey cheating but it works…



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- For F_x, we treat it as a vertical plate:

$$F_x = P_x \times A_{vertical}$$
$$= \rho g h_{CG} A_{vertical}$$

- For F_y, we use the weight of water above:

$$F_y = W$$
$$F_y = \rho g \forall_{above}$$

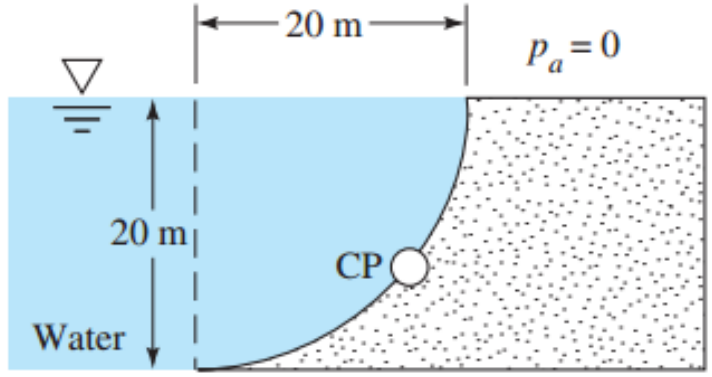
Important note, F_x does not act at h_{CG} it acts at h_{CP}

Link to animations:
https://howardho-cfd.github.io/teaching/fluids/interactive-tools/curved-plate-pressure/

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Problem 2

The dam is a quarter-circle 50 m wide into the paper. Determine the horizontal and vertical components of hydrostatic force against the dam and the point CP where the resultant force strikes the dam.



F_x - Treat as vertical plate

F_y - Weight above

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$

Isothermal: $P = P_1 \exp[-g\Delta z/RT]$

Troposphere: $P = P_0(1 - \beta z/T_0)^{g/R\beta}$ ($\beta = 0.0065$ K/m)

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Curved: $F_H = \gamma h_{CG} A_{proj}$; F_V = Weight of fluid above

Buoyancy: $F_B = \rho g \forall_{sub}$

Stability: $\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$ (Stable if $\overline{MG} > 0$)

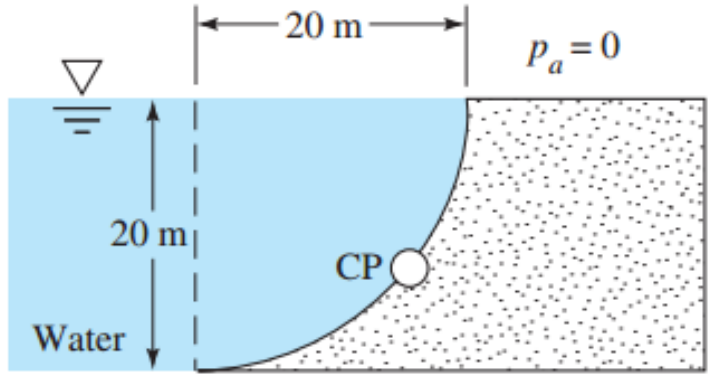
Rigid Body: $\nabla P = \rho(\vec{g} - \vec{a})$; **Rotate:** $z = \frac{\omega^2 r^2}{2g} + C$

Moving manometer: $A_1 a = A_2 b$; inclined-leg vertical change $= b\sin\theta$; write initial/final pressure walks and subtract.

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Problem 2



The dam is a quarter-circle 50 m wide into the paper. Determine the horizontal and vertical components of hydrostatic force against the dam and the point CP where the resultant force strikes the dam.

Start with the easier one first, horizontal pressure force F_x

$$F_x = \rho g h_{CG} A_{vertical}$$

$$= \rho g h_{CG} \, (height \times width)$$

$$= (9790)(10)(20 \times 50)$$

$$= 97.9 \times 10^6 \, N$$

Then the slightly harder vertical pressure force F_y

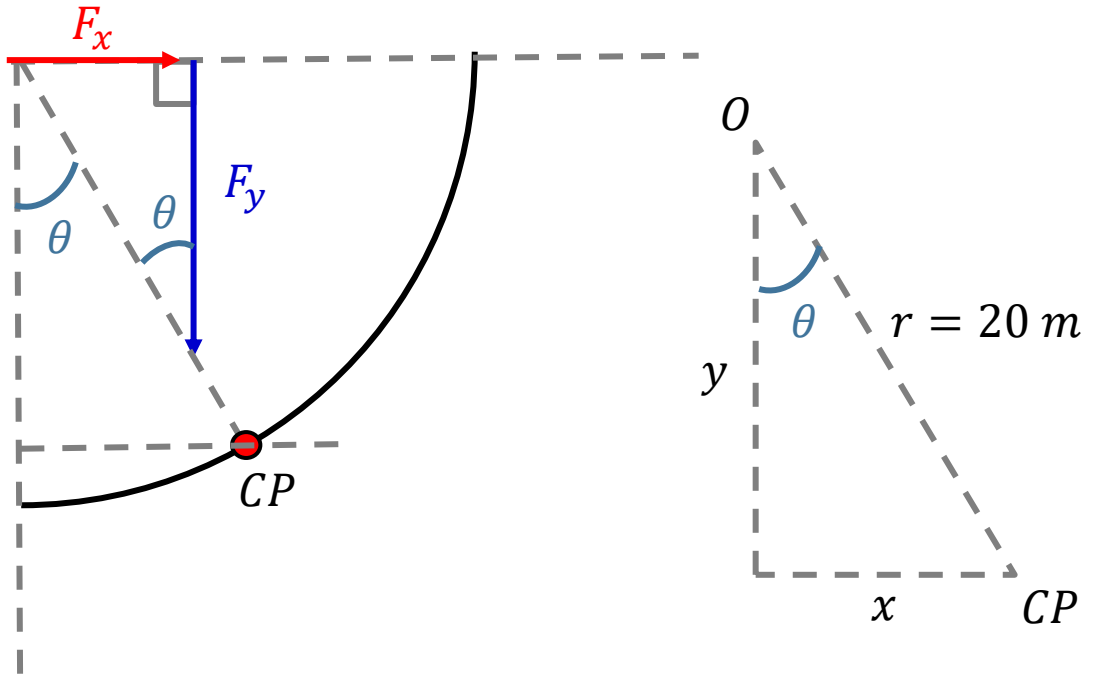
$$F_y = \rho g \forall_{above}$$

$$= \rho g \left(\frac{\pi r^2}{4} \, width \right)$$

$$= (9790) \frac{\pi}{4} (20)^2 (50)$$

$$= 154 \times 10^6 \, N$$

Note: This question is asking for CP as a magnitude of from both F_x and F_y, if we use the h_{CP} equation from the crib sheet it won't work



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$$\tan\theta = \frac{F_x}{F_y}$$

$$\theta = \tan^{-1}\frac{97.9 \times 10^6}{154 \times 10^6}$$

This gives us $\theta = 32.4^o$

Using θ *we can find* y *and* x *since we know* r

$$\sin\theta = \frac{x}{r}$$

$$x = 20\sin 32.4^o$$

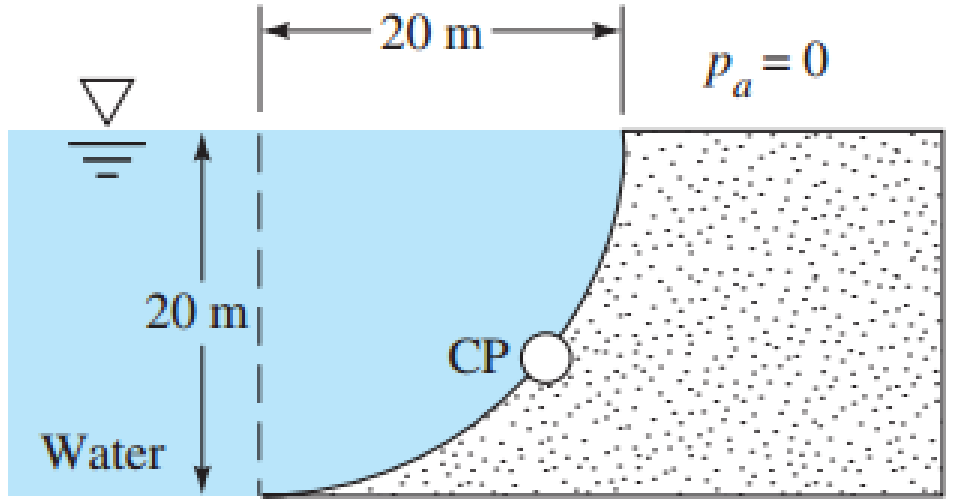
$$x = 10.7\ m$$

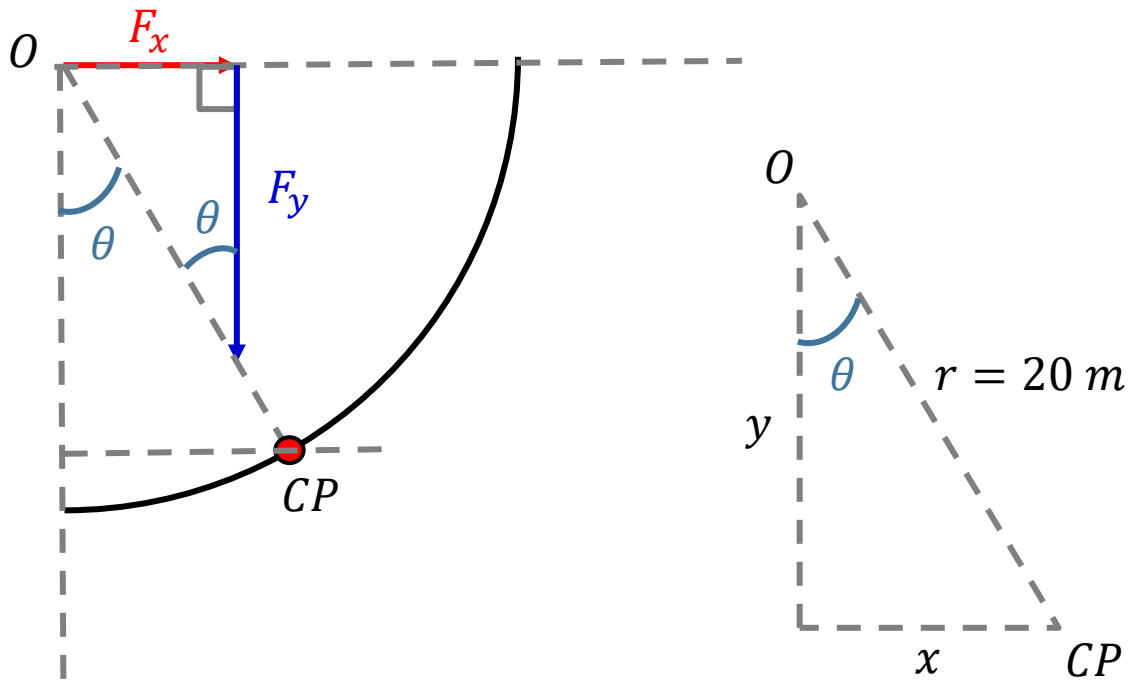
$$\cos\theta = \frac{y}{r}$$

$$y = 20\sin 32.4^o$$

$$y = 16.9\ m$$

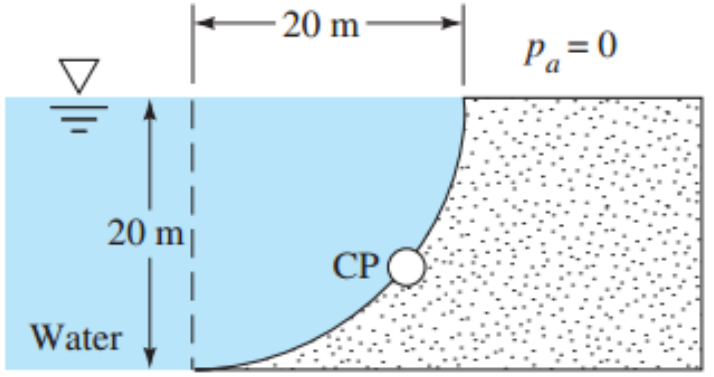
CP is located 10.7 m to the right and 16.9m deep from O





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Problem 2



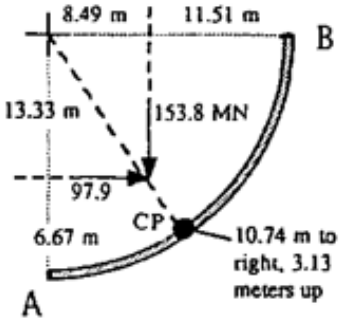
The dam is a quarter-circle 50 m wide into the paper. Determine the horizontal and vertical components of hydrostatic force against the dam and the point CP where the resultant force strikes the dam.

Solution: The horizontal force acts as if the dam were vertical and 20 m high:

$$F_H = \gamma h_{CG} A_{vert}$$
$$= (9790\ \text{N/m}^3)(10\ \text{m})(20 \times 50\ \text{m}^2)$$
$$= \mathbf{97.9\ MN} \quad \textit{Ans.}$$

F_x - Treat as vertical plate
F_y - Weight above

$$P = \rho g h_{CG}$$
$$F_P = P \times A$$



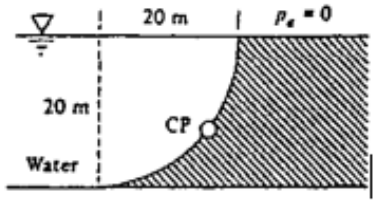


Fig. P2.82

This force acts 2/3 of the way down or 13.33 m from the surface, as in the figure. The vertical force is the weight of the fluid above the dam:

$$F_V = \gamma(\text{Vol})_{dam} = (9790\ \text{N/m}^3)\frac{\pi}{4}(20\ \text{m})^2(50\ \text{m}) = \mathbf{153.8\ MN} \quad \textit{Ans.}$$

This vertical component acts through the centroid of the water above the dam, or $4R/3\pi = 4(20\ \text{m})/3\pi = 8.49$ m to the right of point A, as shown in the figure. The resultant hydrostatic force is $F = [(97.9\ \text{MN})^2 + (153.8\ \text{MN})^2]^{1/2} = \mathbf{182.3\ MN}$ acting down at an angle of **32.5°** from the vertical. The line of action of F strikes the circular-arc dam AB at the center of pressure CP, which is **10.74 m to the right and 3.13 m up from point A**, as shown in the figure. *Ans.*

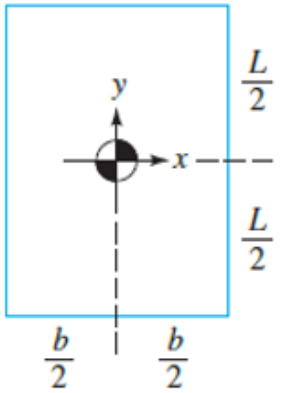
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From Tutorial 3
Angled Plates

Mechanical & Industrial Engineering
UNIVERSITY OF TORONTO

- Pressure force F_P no longer act through Centre of Gravity (CG)
- Centre of Pressure (CP) = location of pressure force F_P
- y_{CP} is distance from CG to CP
- As θ increases, y_{CP} also increases

4



$A = bL$

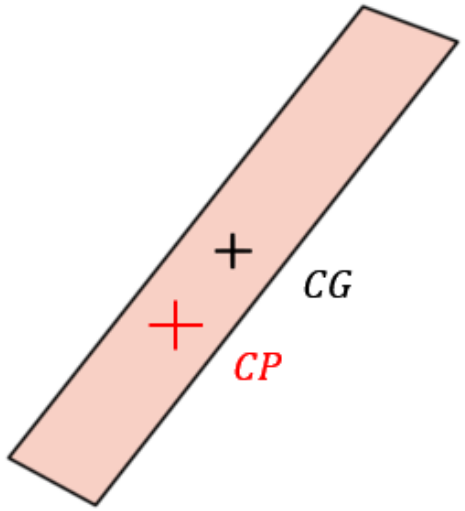
$I_{xx} = \frac{bL^3}{12}$

$I_{xy} = 0$

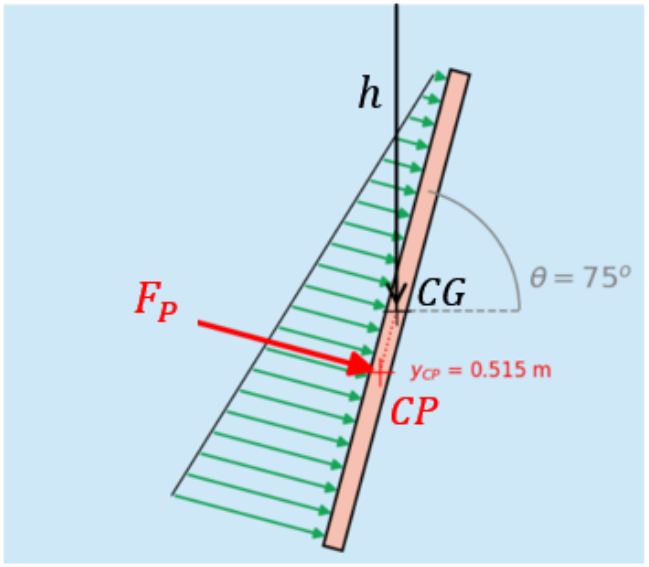
6

$$F = \gamma h_{CG} A \qquad y_{CP} = +\frac{I_{xx}\sin\theta}{h_{CG}A} \qquad x_{CP} = -\frac{I_{xy}\sin\theta}{h_{CG}A}$$

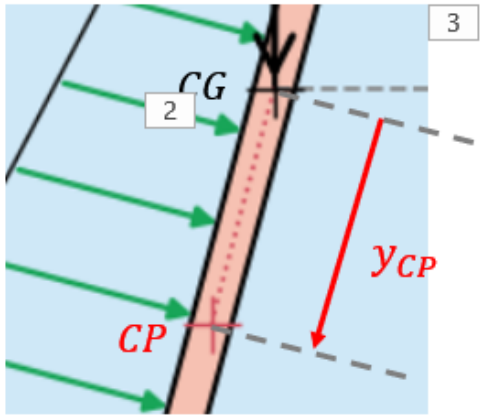
Isometric view



Side View



2 3



The y axis is not vertical, y axis is parallel to the plate !

3



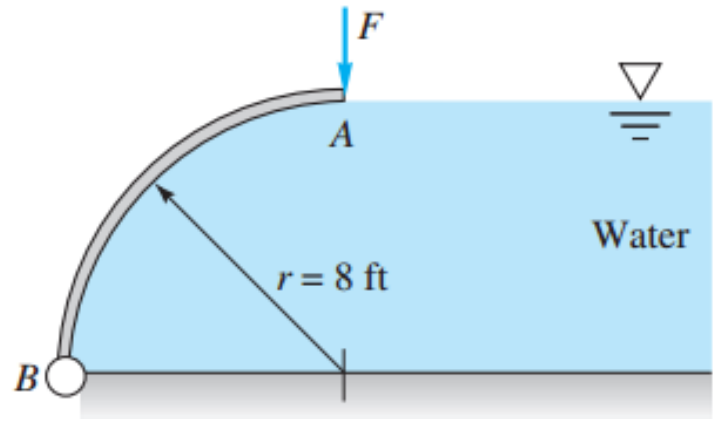
Important Note: The textbook y_{CP} assumes y as upward +ve, we omit the –ve sign for y_{CP} with this coordination system.

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Problem 3

Gate AB in Fig. P2.83 is a quarter circle 10 ft wide into the paper and hinged at B. Find the force F just sufficient to keep the gate from opening. The gate is uniform and weighs 3000 lbf.



b = width

14. GEOMETRIC PROPERTIES

Shape	Area	Centroid	Inertia (I_{xc})
Rectangle	bh	$h/2$	$bh^3/12$
Circle	πR^2	R	$\pi R^4/4$
Semi-Circ	$\pi R^2/2$	$4R/3\pi$	$0.1098R^4$
Quart-Circ	$\pi R^2/4$	$4R/3\pi$	$0.0549R^4$
Triangle	$bh/2$	$h/3$	$bh^3/36$

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$

Isothermal: $P = P_1 \exp[-g\Delta z/RT]$

Troposphere: $P = P_0(1 - \beta z/T_0)^{g/R\beta}$ ($\beta = 0.0065$ K/m)

Hydrostatic: $P = P_0 + \rho g h$

Pascal Law: $F = P \times A$; $P_1 = P_2$

Planes: $F = \gamma h_{CG} A$; $y_{CP} = \frac{I_{xx}\sin\theta}{h_{CG}A}$

Curved: $F_H = \gamma h_{CG} A_{proj}$; F_V = Weight of fluid above

Buoyancy: $F_B = \rho g \forall_{sub}$

Stability: $\overline{MG} = \frac{I_o}{\forall_{sub}} - \overline{GB}$ (Stable if $\overline{MG} > 0$)

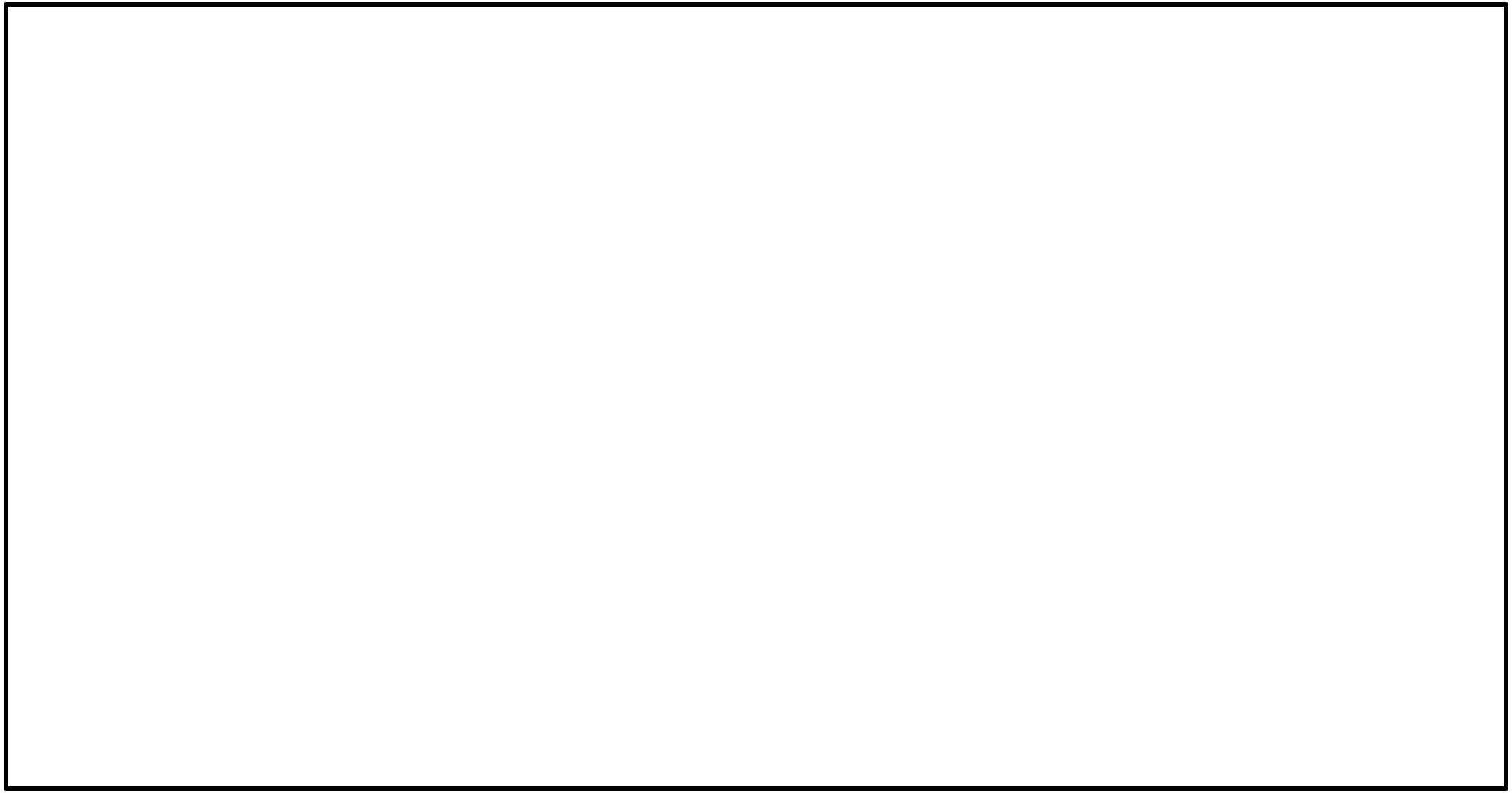
Rigid Body: $\nabla P = \rho(\vec{g} - \vec{a})$; **Rotate:** $z = \frac{\omega^2 r^2}{2g} + C$

Moving manometer: $A_1 a = A_2 b$; inclined-leg vertical change $= b\sin\theta$; write initial/final pressure walks and subtract.

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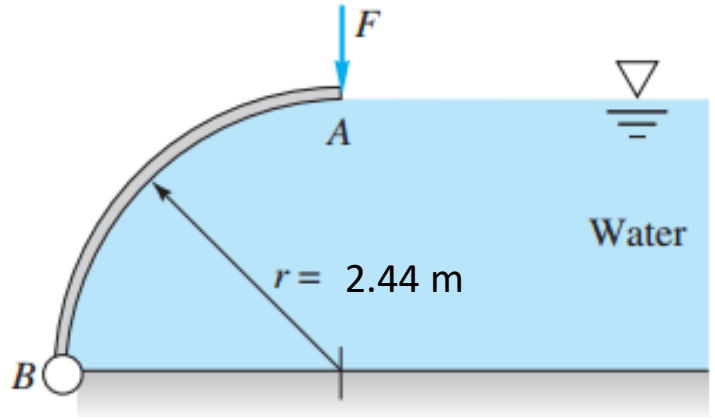


[Link to animations:
https://howardho-cfd.github.io/teaching/fluids/interactive-tools/curved-plate-pressure/](https://howardho-cfd.github.io/teaching/fluids/interactive-tools/curved-plate-pressure/)

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Problem 3

Gate AB in Fig. P2.83 is a quarter circle 3.05 m wide into the paper and hinged at B. Find the force F just sufficient to keep the gate from opening. The gate is uniform and weighs 13.35 kN.



Start with the easier one first, horizontal pressure force F_x

$$F_x = \rho g h_{CG} A_{vertical}$$

$$= \rho g h_{CG} \, (height \times width)$$

$$= (1000)(9.81)(2.44/2)(2.44 \times 3.05)$$

$$= 89067 \, N$$

Naturally first thing I did was convert it to metric

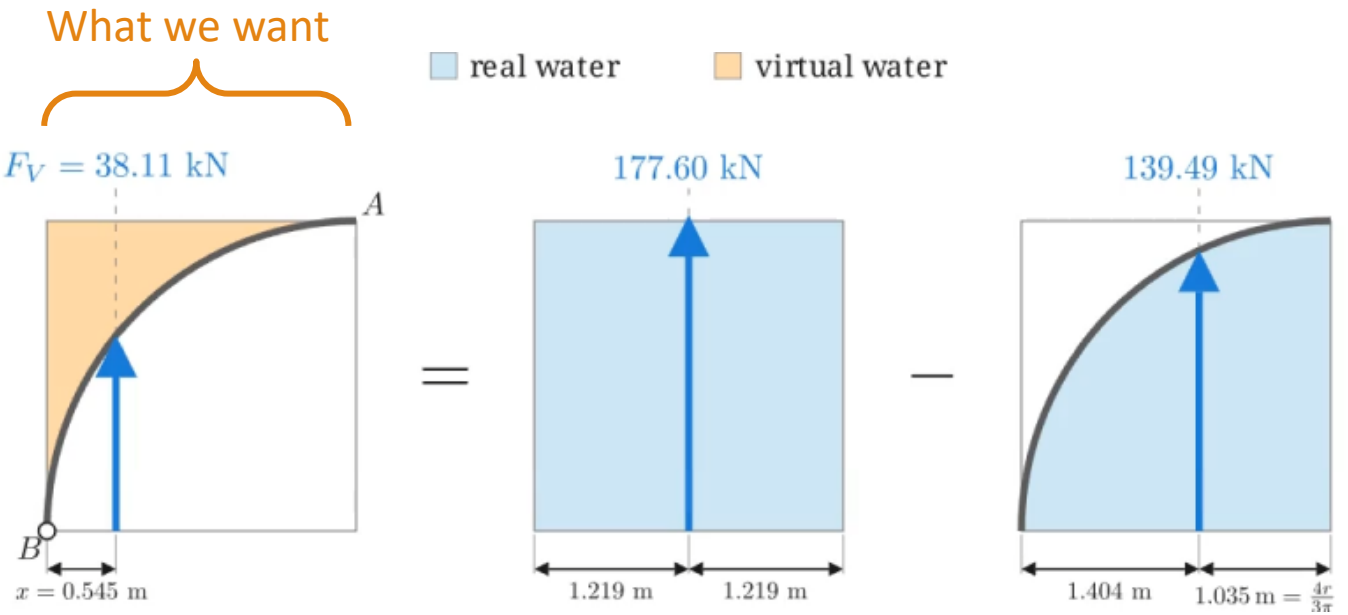
Now the slightly harder vertical pressure force F_y

$$F_y = \rho g \forall_{above}$$

$$= \rho g \, width \, (A_{square} - A_{quarter\ circle})$$

$$= (1000)(9.81)(3.05)\left(2.44^2 - \frac{\pi(2.44^2)}{4}\right)$$

$$= 38228 \, N$$

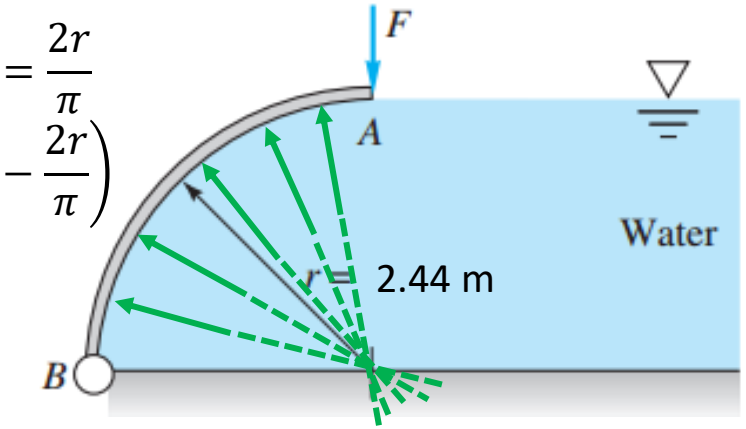


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Method 1 (F_R)

$$x_{gate,CG} = \frac{2r}{\pi}$$

$$x_1 = \left(r - \frac{2r}{\pi}\right)$$



Find magnitude of pressure forces F_R

$$F_R = \sqrt{F_y^2 + F_x^2} = 96924\ N$$

Then draw FBD and take moment at B, (x component of F_R does not contribute here)

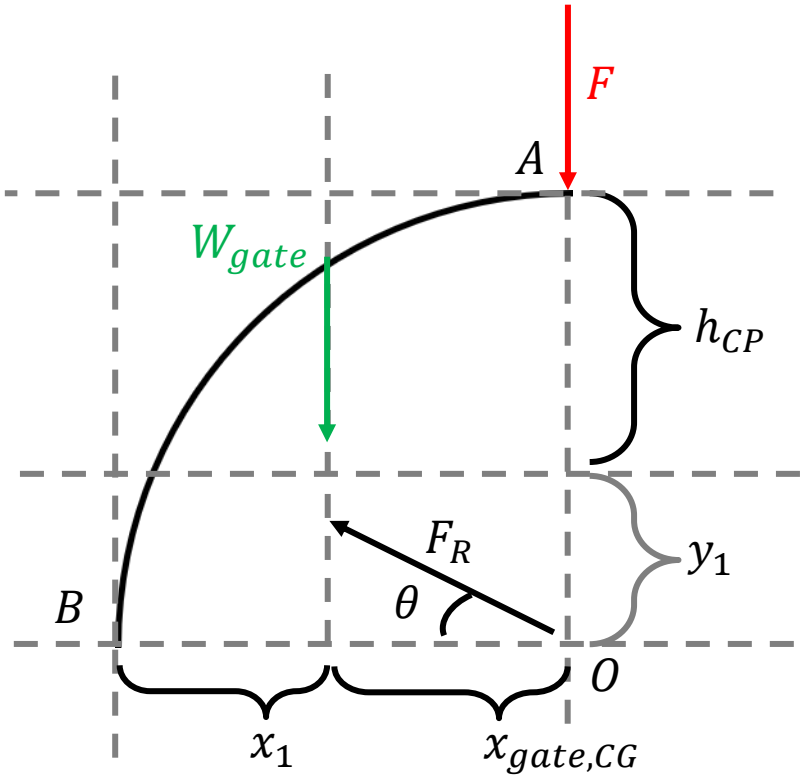
$$\Sigma M_B(CW) = x_1 W_{gate} + rF - rF_R \sin\theta$$

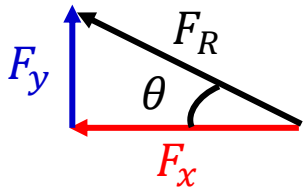
$$0 = \left(2.44 - \frac{2(2.44)}{\pi}\right)(13.35)(1000) + 2.44(F) - 2.44(96924)\sin 23.22^o$$

$$0 = 11837 + 2.44F - 93241$$

$$F = 33362\ N$$

$$F = 7507\ lbf$$





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Method 2 (F_x & F_y)

Find y_{CP} using the h_{CP} equation,

$$h_{CP} = h_{CG} + \frac{I_{xx} \sin\theta}{h_{CG} A}$$

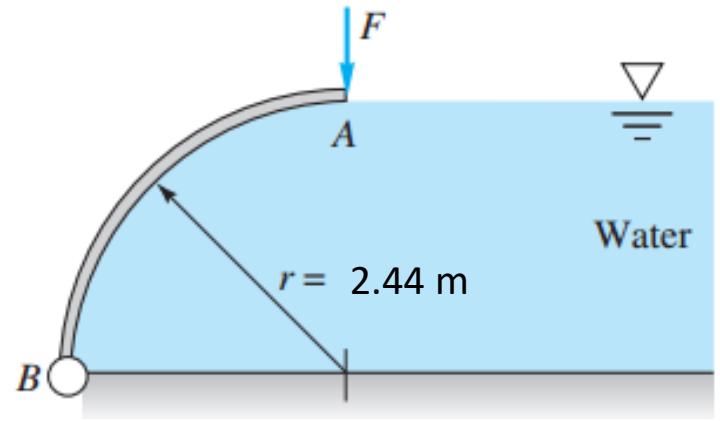
$$h_{CP} = \frac{r}{2} + \frac{(bh^3/12)\sin 90^o}{(r/2)bh}$$

$$h_{CP} = \frac{r}{2} + \frac{h^2 \sin 90^o}{6r}$$

$$h_{CP} = \frac{2.44}{2} + \frac{2.44^2 \sin 90^o}{6(2.44)}$$

$$h_{CP} = 1.63\ m$$

h_{CP} is located 1.63 *m below A or* **0.81 *m above B (we define $y_1 = 0.81\ m$)***



b = width

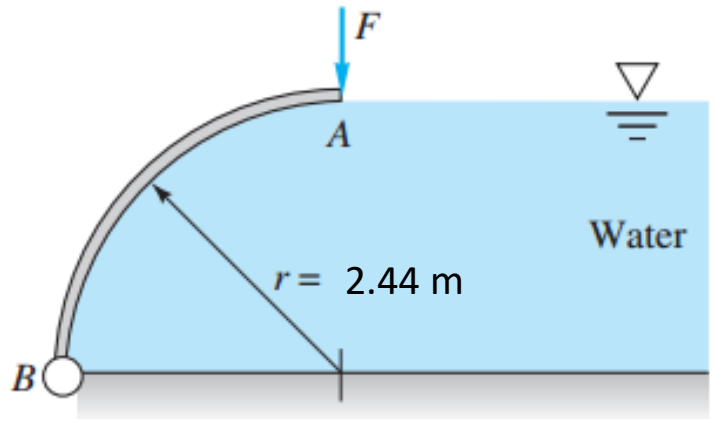
14. GEOMETRIC PROPERTIES

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Quart-Circ	$\pi R^2/4$	$4R/3\pi$	$0.0549R^4$
Triangle	$bh/2$	$h/3$	$bh^3/36$

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Pascal Law: $F = P \times A$; $P_1 = P_2$
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Rigid Body: $\nabla P = \rho(\vec{g} - \vec{a})$; **Rotate:** $z = \frac{\omega^2 r^2}{2g} + C$
Moving manometer: $A_1 a = A_2 b$; inclined-leg vertical change $= b\sin\theta$; write initial/final pressure walks and subtract.

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Now, let's draw our FBD and construct the moment equation from B (we are talking moment at B so we don't have to figure out the normal reaction force there)

$$\Sigma M_B(CW) = r \times F + x_1 \times W_{gate} - y_1 \times F_x - \boldsymbol{x_2 \times F_y}$$

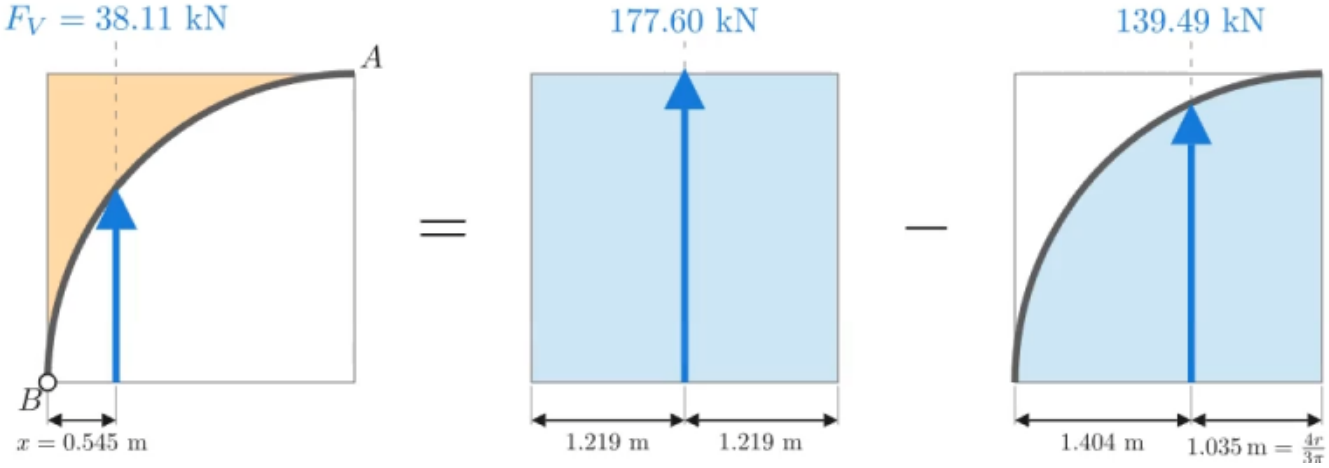
But at this stage, we have too many unknowns, we need to figure out where F_y is acting on. To do that we to go back to the virtual water diagrams that we use to find F_y and calculate the moment caused by the virtual water

$$moment_{virtual\ water} = moment_{square} - moment_{quarter\ circle}$$

$$x_2 \times F_y = (r/2) \times F_{y,square} - x_{qc,B} \times F_{y,quarter\ circle}$$

(We are basically using the area subtraction method again but for moment, it's really weird but it works somehow)

This gives $x_2 = 0.545\ m$



Centroid of quarter circle

$x_{qc,O} = \frac{4r}{3\pi}$ (from O)

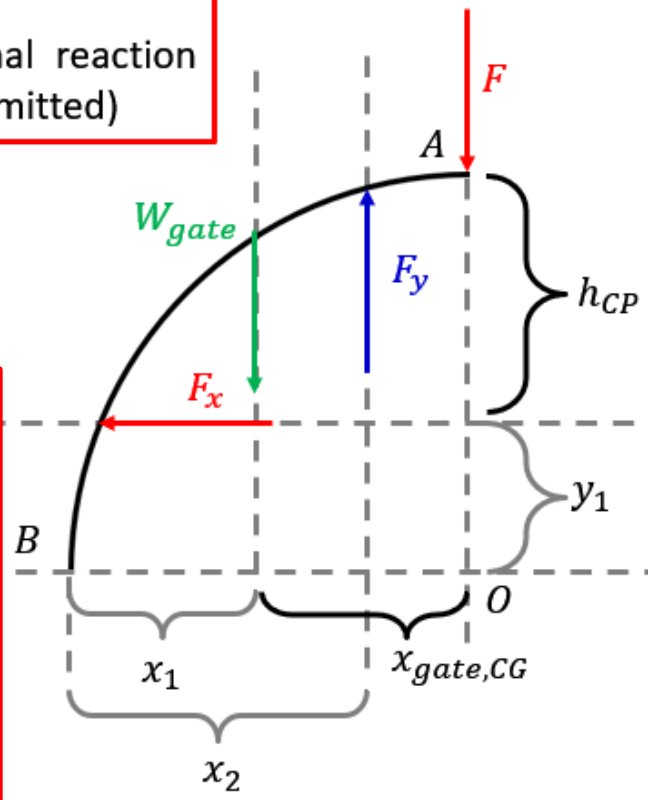
$x_{qc,B} = r - \frac{4r}{3\pi}$ (from B)

Centroid of arc

$x_{gate,CG} = \frac{2r}{\pi}$ (from O)

$x_1 = \left(r - \frac{2r}{\pi}\right)$ (from B)

FBD
(Normal reaction at B omitted)



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Finally, we can sub. everything into the ΣM_B equation and solve for holding force F.

$$\Sigma M_B(CW) = r \times F + x_1 \times W_{gate} - y_1 \times F_x - x_2 \times F_y$$

Sum of moment has to be zero for curved plate to be stationary (LHS=0),

$$0 = r \times F + x_1 \times W_{gate} - y_1 \times F_x - x_2 \times F_y$$

$$0 = 2.44\,F + \left(2.44 - \frac{2(2.44)}{\pi}\right)(13.35)(1000) - 0.81\,(89067) - 0.545(38228)$$

$$F = 33254\ N$$

$$\boxed{F = 7482\ lbf}$$

$\Sigma M_B(\text{clockwise}) = 0 = F(8.0) + (3000)(2.907) - (8570)(1.787) - (19968)(2.667),$

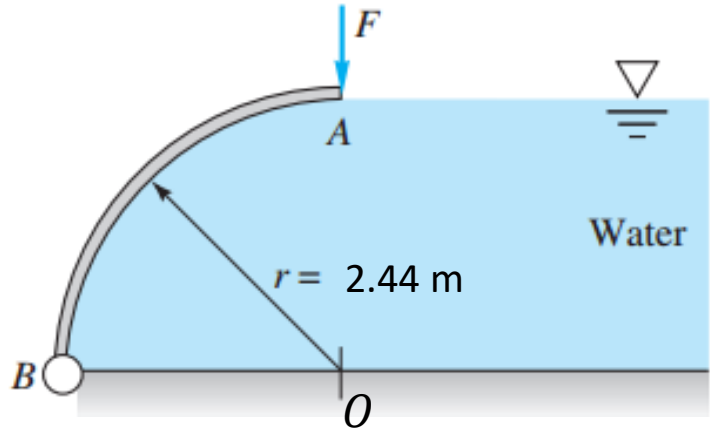
or $F = \frac{59840}{8.0} = \mathbf{7480\ lbf}$ *Ans.*

$1\ N \approx 0.225\ lbf \quad 1m = 3.28\ ft$

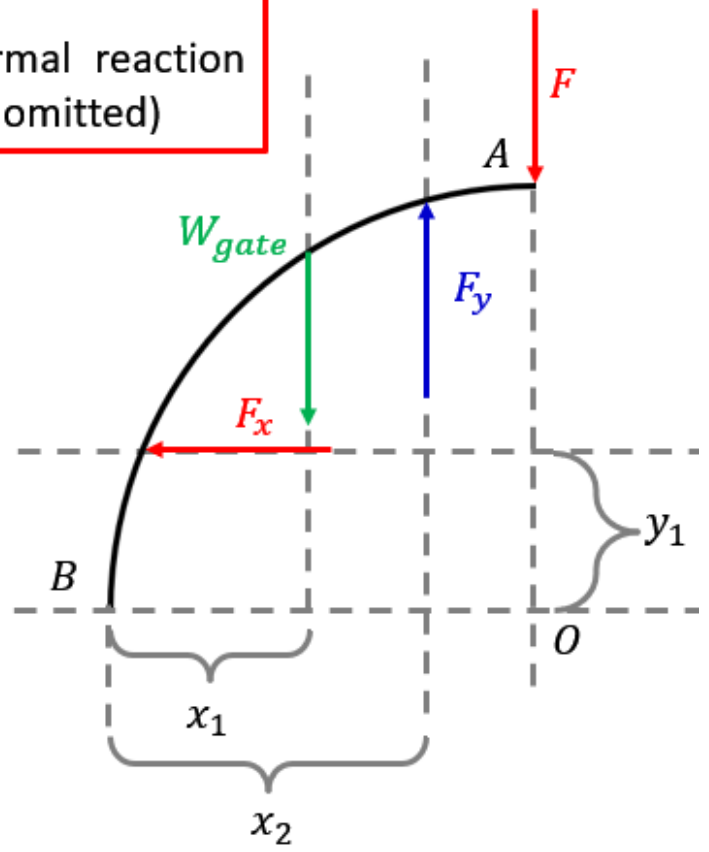
From previous calculations,

$x_2 = 0.545\ m \qquad y_1 = 0.81\ m$

$F_x = 89067\ N \qquad F_y = 38228\ N$



FBD
(Normal reaction at B omitted)



Centroid of quarter circle

$x_{qc} = \frac{4r}{3\pi}$

Centroid of arc (from O)

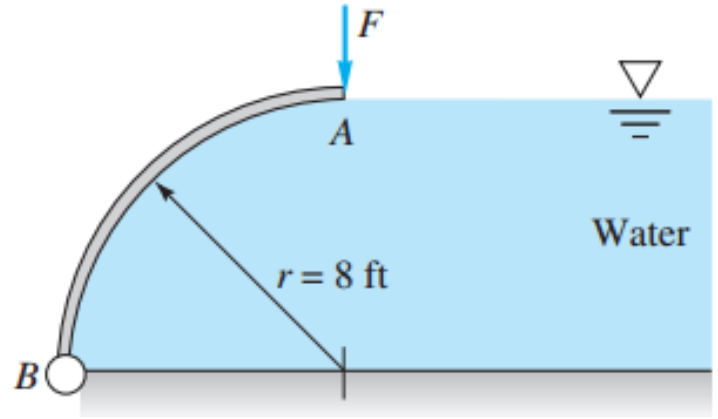
$x_{gate,CG} = \frac{2r}{\pi}$

$x_1 = \left(r - \frac{2r}{\pi}\right)$

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Problem 3

Gate AB in Fig. P2.83 is a quarter circle 10 ft wide into the paper and hinged at B. Find the force F just sufficient to keep the gate from opening. The gate is uniform and weighs 3000 lbf.

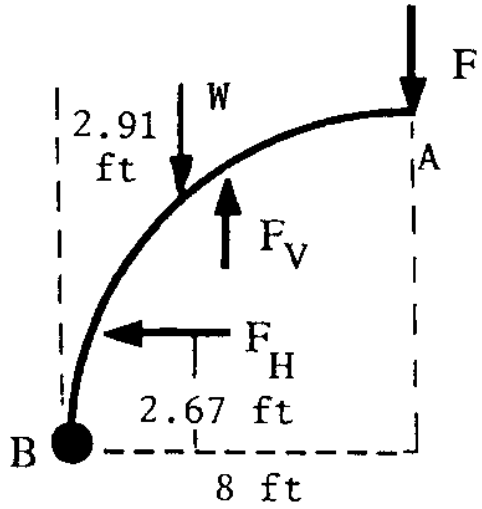


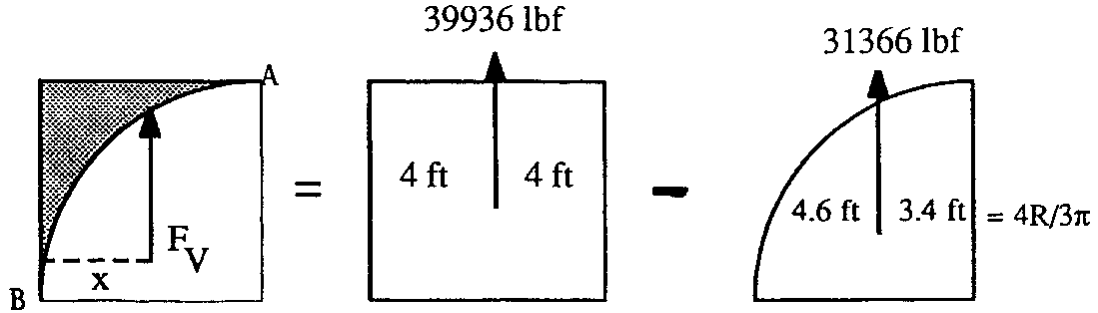
Solution: The horizontal force is computed as if AB were vertical:

$$F_H = \gamma h_{CG} A_{vert} = (62.4)(4\text{ ft})(8 \times 10\text{ ft}^2)$$
$$= 19968\text{ lbf} \quad \text{acting } 5.33\text{ ft below A}$$

The vertical force equals the weight of the missing piece of water above the gate, as shown below.

$$F_V = (62.4)(8)(8 \times 10) - (62.4)(\pi/4)(8)^2(10)$$
$$= 39936 - 31366 = 8570\text{ lbf}$$





The line of action x for this 8570-lbf force is found by summing moments from above:

$$\sum M_B(\text{of } F_V) = 8570x = 39936(4.0) - 31366(4.605), \quad \text{or} \quad x = 1.787\text{ ft}$$

This problem gave myself and some of your upper year peers quite a bit of headache back then, it's brutal already and the solution is in imperial units (cheery on top of a cake…)

$$1\ N \approx 0.225\ lbf \qquad 1m = 3.28\ ft$$

Finally, there is the 3000-lbf gate weight W, whose centroid is $2R/\pi = 5.093$ ft from force F, or $8.0 - 5.093 = 2.907$ ft from point B. Then we may sum moments about hinge B to find the force F, using the freebody of the gate as sketched at the top-right of this page:

$$\sum M_B(\text{clockwise}) = 0 = F(8.0) + (3000)(2.907) - (8570)(1.787) - (19968)(2.667),$$

$$\text{or} \quad F = \frac{59840}{8.0} = \mathbf{7480\ lbf} \quad \textit{Ans.}$$

The key to finding x is by subtracting the moments of the square and the quarter circle

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